Charles "Chaz" Merritt

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Summary

Geospatial and forest-modeling analyst with an M.S. in Artificial Intelligence and experience automating raster/vector workflows in Python, developing landscape-scale decision models, and publishing applied forestry research.

Education

University of Georgia, MS in Artificial Intelligence — 2024 – 2026

- **Coursework:** Machine Learning, Game Theory, Advanced Forest Planning
- **Thesis:** Stochastic Valuation of Planted Loblolly Pine Stands Using Monte Carlo Simulation

University of Georgia, BA in Cognitive Science — 2020 – 2024

- **Coursework:** Search & Optimization, Bayesian Statistics, Epistemology

Experience

Research Professional, University of Georgia – Athens, GA — May 2026 – Present

- Developed a landscape-scale timber modeling and optimization framework integrating LANDFIRE, TreeMap, and FIA to support forest management for stakeholders across the Southeast US.
- Co-authored peer-reviewed papers on AI and forestry.

Research Assistant, University of Georgia – Athens, GA — Aug. 2024 – May 2026

- Built Monte Carlo forest-valuation models using CVaR to quantify uncertainty and support risk-informed management.

Geospatial Analyst Intern, Northwest Management – Moscow, ID — May – Sep. 2024

- Automated raster and vector GIS pipelines for forest stand delineation using Python, Rasterio, GeoPandas, leading to a 3x speedup in delineation procedures.
- Utilized clustering algorithms and image segmentation models (SAM) for forest analysis.

Selected Projects

Spatial Equilibrium Timber Supply Web Interface — 2026

- Built a stakeholder-facing interface to a Pyomo-based LP timber-supply model, translating complex outputs for nontechnical users.
- Used a strangler pattern to refactor the legacy file-based system into a DataFrame based monorepo leading to massive performance gains.

Brandon AI — 2026

- Developed contrastive representation models for phonetic similarity learning with InfoNCE loss.
- Gained experience working in a startup environment and led benchmarking of the product.

Selected Publications

Tree Species Classification from Smartphone Bark Images with DiNOv3 Embeddings*, Pending* — July. 2026
Chaz Merritt, Krista Merry, Pete Bettinger, Roger C. Lowe III

AI Technology Use in Forestry, *Journal of Forestry* — Feb. 2026
Krista Merry, Pete Bettinger, ***Chaz Merritt***, Khaled Rasheed, Bruno da Silva, Roger C. Lowe III